

A Zero-shot Generalized Graph Anomaly Detection Framework via Node Reconstruction

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Abstract. Cross-domain graph anomaly detection (GAD) aims to identify abnormal nodes in unseen target graphs, showing strong potential in real-world applications with heterogeneous graph data. However, existing methods often depend on dataset-specific feature semantics and structural patterns, which limits their ability to generalize across different domains. To address this challenge, we propose AlignGAD, a zero-shot generalized graph anomaly detection framework. Our framework is built upon three key components: a Global Unification Module that aligns heterogeneous node features and normalizes graph signals in the spectral domain; a Clustering Module that constructs cluster-aware graph views to capture group-level abnormal patterns; and a Node Discrepancy Scoring Module that measures reconstruction discrepancy and aggregates anomaly evidence from different graph views. Experiments on multiple real-world datasets demonstrate the effectiveness of AlignGAD under the zero-shot GAD setting.

Keywords: Zero-shot Learning · Graph Anomaly Detection · Information Unification · Clustering · Generalized Framework.

1 Introduction

Graph Anomaly Detection (GAD) aims to identify nodes that deviate from normal patterns in graph-structured data. It has become important in many real-world applications, including multimedia security [10, 12], social network analysis [9, 19], and intelligent surveillance [29, 30]. However, most existing GAD methods are designed for a specific graph or domain. They often depend on domain-specific structural and semantic patterns [18], which makes them difficult to transfer to unseen graphs with different feature spaces, graph structures, and anomaly behaviors [2].

Recently, generalized GAD has attracted increasing attention as a more practical setting. Instead of training a separate detector for each dataset, the goal is to train one model on source graphs and directly apply it to unseen target graphs. Some recent studies have explored this direction, such as ARC [14], but generalizing across heterogeneous graphs remains challenging. First, graphs from different domains may have different feature dimensions and semantic meanings,

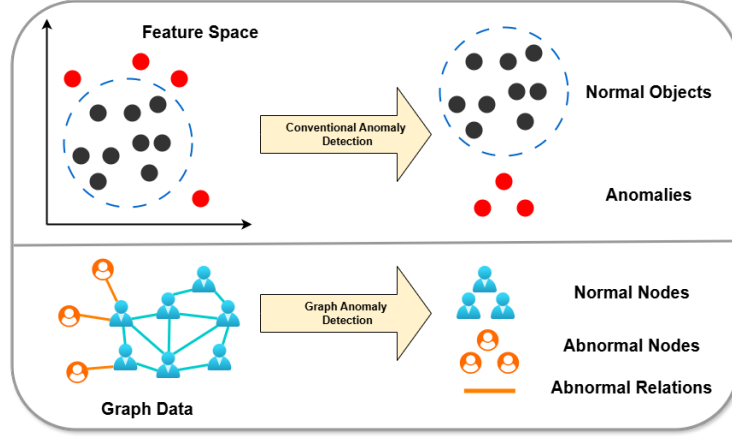


Fig. 1. Difference between conventional anomaly detection and graph anomaly detection. Conventional anomaly detection focuses on isolated objects in feature space, while graph anomaly detection considers both node attributes and graph relations to identify abnormal nodes and abnormal connections.

making direct feature transfer unreliable. Second, anomalies are not always expressed at the individual node level. In many cases, a node may look less suspicious alone, but become more informative when considered together with the cluster or group it belongs to.

To address these challenges, we propose AlignGAD, a zero-shot framework for graph anomaly detection across heterogeneous graphs. AlignGAD first uses a Global Unification Module to align node feature dimensions and normalize graph signals in the spectral domain. It then introduces a Clustering Module to build cluster-aware graph views, allowing the model to capture group-level abnormal patterns that support node-level detection. Finally, a Node Discrepancy Scoring Module reconstructs node features and measures the discrepancy between input and generated representations. The final anomaly score is obtained by aggregating evidence from both node-level and cluster-level views. Our main contributions are summarized as follows:

- We propose AlignGAD, a zero-shot GAD framework that learns from source graphs and directly generalizes to unseen graphs without fine-tuning. By aligning graph features and spectral distributions, AlignGAD reduces the dependence on domain-specific semantics.
- We introduce a cluster-aware discrepancy scoring strategy that combines node reconstruction with cluster-level graph views. This allows the model to capture both individual node deviations and group-level abnormal patterns.
- Extensive experiments on multiple real-world graph datasets demonstrate the effectiveness of AlignGAD under the zero-shot setting. The results show that our framework provides a practical and generalizable solution for anomaly detection across heterogeneous graphs.

2 Related Works

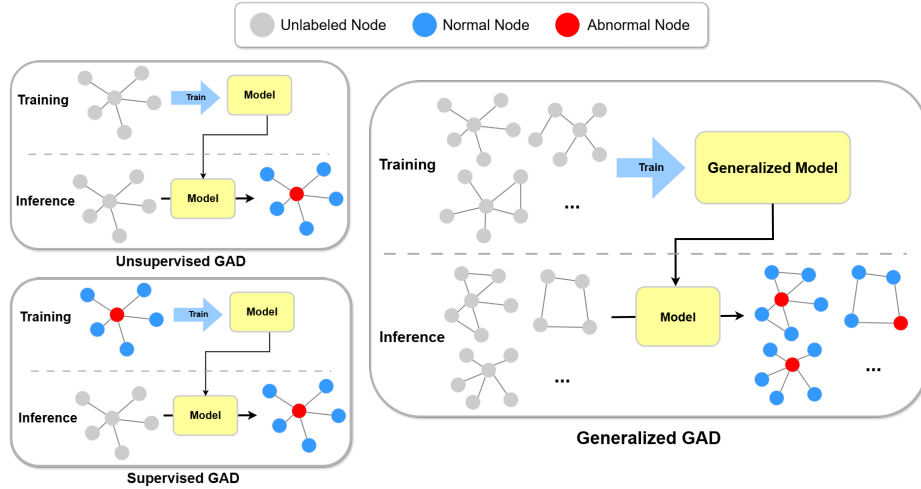


Fig. 2. Comparison of GAD settings. Conventional supervised and unsupervised GAD methods train and infer within the same domain, while generalized GAD trains on source graphs and directly detects anomalies in unseen target graphs.

Graph Anomaly Detection. Graph Anomaly Detection (GAD) aims to identify nodes or local graph structures that exhibit abnormal behaviors compared to the majority of the graph [23, 16, 8]. Existing approaches are generally categorized according to the level of supervision available during training. Supervised methods, such as BGNN [11], BWGNN [23], GHRN [7], and CAGAD [27], learn discriminative decision boundaries from labeled normal and anomalous nodes, but their applicability is often limited by the scarcity of anomaly annotations. Semi-supervised approaches alleviate this issue by assuming that only a subset of normal nodes is available, as demonstrated by S-GAD [21], which synthesizes artificial outliers from normal samples to train a one-class classifier. In contrast, unsupervised methods operate without any labels and typically rely on reconstruction or representation learning objectives. For example, DOMINANT [4] employs a graph autoencoder to reconstruct graph topology and node attributes, using reconstruction errors as anomaly scores, while CoLA [15] leverages contrastive learning to capture discrepancies between target nodes and their neighborhood subgraphs. Methods such as TAM [20] and GCTAM [28] further extend one-class learning paradigms to graph data by optimizing affinity-based objectives in the latent space. Although these methods achieve strong performance on individual datasets, they generally require retraining for each new graph, limiting their ability to generalize across domains.

Generalist Graph Anomaly Detection. To overcome the limited transferability of conventional GAD models, recent studies have explored generalist graph anomaly detection frameworks that seek to learn reusable anomaly knowledge across diverse graph domains. While some cross-domain approaches [5, 25] can adapt to unseen datasets, they often depend on strong correlations between source and target graphs, such as aligned feature spaces or consistent semantic meanings. ARC [14] introduces a unified GAD framework based on in-context learning, where node representations from different graphs are projected into a shared latent space, residual neighborhood information is encoded through an ego-neighbor graph encoder, and anomaly scores are computed via cross-attention against a small set of normal reference samples. This design enables adaptation to new graphs with minimal additional supervision. More recently, UNPrompt [17] proposes a zero-shot generalist GAD framework trained on a single source graph. By performing coordinate-wise normalization across node attributes and learning transferable neighborhood prompts, UNPrompt estimates anomaly scores through the predictability of latent node representations. Despite their promising results, both methods exhibit notable limitations: ARC still requires a few normal target-domain samples during inference and may not fully eliminate domain shifts, whereas UNPrompt relies on consistent attribute semantics and a single-source training paradigm, which can reduce its robustness when graph structures and feature distributions vary substantially across domains.

3 Preliminary

Notations. The structure of i -th graph is defined as $\mathcal{G}_i = (\mathcal{V}_i, \mathcal{E}_i, A_i, X_i)$. Here, $\mathcal{V}_i = \{v_1, \dots, v_{N_i}\}$ is the set of nodes with size $N_i = |\mathcal{V}_i|$, and $\mathcal{E}_i \subseteq \mathcal{V}_i \times \mathcal{V}_i$ represents the edge set. The structural information is captured by the adjacency matrix $A_i \in \{0, 1\}^{N_i \times N_i}$. The node feature matrix $X_i \in \mathbb{R}^{N_i \times d_i}$ encodes node attributes, with d_i denoting the feature dimension.

Conventional GAD. Given a graph $\mathcal{G}_i$, conventional GAD aims to learn a scoring function $f : \mathcal{V}_i \rightarrow \mathbb{R}$ that assigns an anomaly score to each node. Let $\mathcal{V}_i^n$ and $\mathcal{V}_i^a$ denote the normal and anomalous node sets, respectively, where anomalous nodes are usually much fewer than normal nodes, i.e., $|\mathcal{V}_i^a| \ll |\mathcal{V}_i^n|$. The scoring function is expected to give higher scores to anomalous nodes than normal ones, namely $f(v_a) > f(v_n)$ for $v_a \in \mathcal{V}_i^a$ and $v_n \in \mathcal{V}_i^n$. Most conventional GAD methods train and test within the same graph or domain, which makes them sensitive to dataset-specific feature semantics and structural patterns.

Zero-shot GAD. Zero-shot GAD considers a more practical but harder setting, where a model is trained on source graphs $\mathcal{G}_{\text{train}} = \{\mathcal{G}_1, \dots, \mathcal{G}_m\}$ and directly applied to unseen target graphs $\mathcal{G}_{\text{test}} = \{\mathcal{G}_{m+1}, \dots, \mathcal{G}_{m+t}\}$ without further tuning. During inference, no target-domain labels or additional supervision are available. The goal is to learn a general scoring function f_θ that can assign reliable

anomaly scores $s_v = f_\theta(v, \mathcal{G}_j)$ for nodes in any unseen graph $\mathcal{G}_j \in \mathcal{G}_{\text{test}}$. This setting requires the model to capture transferable graph patterns rather than relying on domain-specific semantics.

Graph Fourier Transform. Graph Fourier Transform (GFT) [1] maps graph signals into the spectral domain using the eigendecomposition of the graph Laplacian. For the i -th graph, we define the Laplacian as $L_i = D_i - A_i$, where D_i is the degree matrix and A_i is the adjacency matrix. The eigendecomposition of L_i is given by:

$$L_i = U_i \Lambda_i U_i^\top, \quad 0 = \lambda_1 \leq \dots \leq \lambda_{N_i} \leq 2, \quad (1)$$

where U_i contains the orthonormal eigenvectors and Λ_i is the diagonal matrix of eigenvalues. Given a graph signal $x \in \mathbb{R}^{N_i}$, the GFT and inverse GFT are defined as $\hat{x} = U_i^\top x$ and $x = U_i \hat{x}$, respectively. In the spectral domain, low-frequency components with small λ_k usually describe smooth global patterns, while high-frequency components with large λ_k capture sharper local variations. This relation can be expressed through the signal smoothness:

$$x^\top L_i x = \sum_{k=1}^{N_i} \lambda_k |\hat{x}_k|^2 = \sum_{k=1}^{N_i} \lambda_k E(\lambda_k). \quad (2)$$

A smooth graph signal has most of its energy concentrated in low-frequency components, whereas anomalous or irregular nodes often disturb this pattern by increasing high-frequency energy.

4 Method

In this section, we present AlignGAD, a zero-shot framework for graph anomaly detection across heterogeneous graphs. As shown in Figure 3, AlignGAD is built with three main modules. The Global Unification Module first aligns node features from different graphs and normalizes their spectral representations. The Clustering Module then groups nodes into cluster-aware views, allowing the model to capture group-level abnormal patterns that support node-level anomaly prediction. Finally, the Node Discrepancy Scoring Module reconstructs node features and measures the discrepancy between reconstructed and generated representations to produce anomaly scores. The outputs from different graph views are aggregated to obtain the final prediction.

4.1 Global Unification Module

A central difficulty in zero-shot GAD is that graphs from different domains often have incompatible feature spaces and different spectral distributions. To make these graphs comparable, the Global Unification Module first projects node features into a shared feature dimension using Singular Value Decomposition (SVD). It then applies Graph Fourier Transform and spectral normalization to reduce distribution gaps in the frequency domain. This module provides a unified representation for heterogeneous graphs.

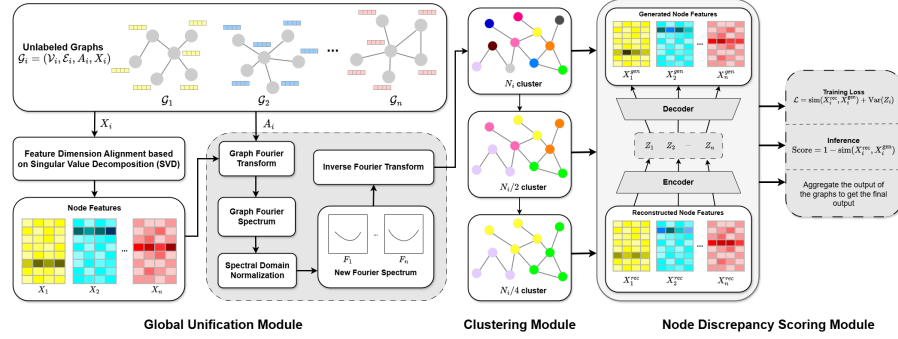


Fig. 3. The overall framework of AlignGAD. The framework consists of three modules: Global Unification Module, Clustering Module, and Node Discrepancy Scoring Module. The anomaly score is obtained by aggregating the outputs from different graph views.

Feature Dimension Alignment. Graphs from different domains often contain node features with inconsistent dimensions, making direct cross-domain learning difficult. Simple operations such as zero-padding or feature truncation may destroy important semantic information and reduce generalization ability. To address this issue, we first align all node features into a shared latent dimension using Singular Value Decomposition (SVD). Given a graph $\mathcal{G}_i$ with node feature matrix $X_i \in \mathbb{R}^{N_i \times d_i}$, we compute

$$X_i = U_i \Sigma_i V_i^\top, \quad (3)$$

where U_i and V_i are orthogonal matrices, and Σ_i contains the singular values. We retain the top- d' singular components to construct a unified representation:

$$X_i^{align} = U_i^{(:,1:d')} \Sigma_i^{(1:d',1:d')} \left(V_i^{(:,1:d')} \right)^\top, \quad (4)$$

where d' denotes the shared feature dimension across all graphs. In this way, the aligned representation preserves the dominant structural information while reducing feature inconsistencies between heterogeneous graphs.

Spectral Domain Normalization. After feature dimension alignment, graphs may still differ in how their signals are distributed over the graph spectrum. Some graphs have smoother signals dominated by low-frequency components, while others contain stronger local variations in high-frequency components. To reduce this spectral gap, we normalize the aligned node features in the frequency domain. For each graph $\mathcal{G}_i$, we first decompose its Laplacian matrix as

$$L_i = U_i \Lambda_i U_i^\top, \quad (5)$$

where U_i is the graph Fourier basis and Λ_i contains the eigenvalues. We then project the aligned features X_i^{align} into the spectral domain:

$$F_i = U_i^\top X_i^{align} \quad (6)$$

To make spectral representations more comparable across graphs, we standardize each frequency component:

$$\hat{F}_i = \frac{F_i - \mu_i}{\sigma_i}, \quad (7)$$

where the mean and standard deviation are computed as

$$\mu_i = \frac{1}{N_i} \mathbf{1}^\top F_i, \quad \sigma_i = \sqrt{\frac{1}{N_i} \mathbf{1}^\top (F_i \odot F_i) - \mu_i \odot \mu_i} \quad (8)$$

Here, $\mathbf{1} \in \mathbb{R}^{N_i}$ is an all-one vector, and $\odot$ denotes element-wise multiplication. The normalized spectral features are finally mapped back to the node space through inverse GFT:

$$X_i^{rec} = U_i \hat{F}_i = U_i \left(\frac{U_i^\top X_i^{align} - \mu_i}{\sigma_i} \right) \quad (9)$$

The resulting representation X_i^{rec} preserves the graph structure while reducing spectral distribution differences across domains. This provides a more stable input for the following clustering and node discrepancy scoring modules.

4.2 Clustering Module

The Clustering Module is designed to provide group-level structural context for node anomaly detection. Instead of only scoring each node in the original graph, we build a set of cluster-aware graph views with progressively coarser structures. Given the unified node representation X_i^{rec} from the Global Unification Module, we first keep the original graph as the finest view, where each node can be regarded as one cluster. We denote this view as $\mathcal{G}_i^{(0)} = (\mathcal{V}_i^{(0)}, \mathcal{E}_i^{(0)}, A_i^{(0)}, X_i^{(0)})$, where $X_i^{(0)} = X_i^{rec}$ and $A_i^{(0)} = A_i$.

For the next views, we apply K-means clustering to group nodes into fewer cluster nodes. Specifically, we construct three graph views with cluster sizes N_i , $N_i/2$, and $N_i/4$. For the r -th clustering view, K-means assigns each node in the previous view to one cluster:

$$\pi_i^{(r)} = \text{KMeans}(X_i^{(r-1)}, K_i^{(r)}), \quad K_i^{(r)} = \left\lfloor \frac{N_i}{2^r} \right\rfloor, \quad r \in \{1, 2\}. \quad (10)$$

Here, $\pi_i^{(r)}$ is the cluster assignment function, and $K_i^{(r)}$ is the number of clusters in the r -th view. Each cluster is then treated as a new node, whose feature is computed by averaging the features of all nodes assigned to that cluster:

$$x_{i,c}^{(r)} = \frac{1}{|\mathcal{C}_{i,c}^{(r)}|} \sum_{v \in \mathcal{C}_{i,c}^{(r)}} x_{i,v}^{(r-1)}, \quad (11)$$

where $\mathcal{C}_{i,c}^{(r)}$ denotes the set of nodes assigned to the c -th cluster in view r . The adjacency matrix of the clustered graph is also updated according to the connections between clusters:

$$A_{i,cd}^{(r)} = \mathbb{I} \left[\exists u \in \mathcal{C}_{i,c}^{(r)}, \exists v \in \mathcal{C}_{i,d}^{(r)} \text{ such that } A_{i,uv}^{(r-1)} = 1 \right]. \quad (12)$$

In this way, the second view is built from the nodes in the first view, and the third view is built from the cluster nodes in the second view. These progressively coarsened graphs allow the model to observe whether a node is suspicious not only by itself, but also through the behavior of the group it belongs to. The resulting graph views are then passed to the Node Discrepancy Scoring Module for anomaly scoring.

4.3 Node Discrepancy Scoring Module

The Node Discrepancy Scoring Module measures how much a node deviates from the representation generated by a graph autoencoder. After the Clustering Module, each graph $\mathcal{G}_i$ is represented by three graph views, denoted as $\{\mathcal{G}_i^{(r)}\}_{r=0}^2$. Here, $\mathcal{G}_i^{(0)}$ is the finest view with N_i nodes, while $\mathcal{G}_i^{(1)}$ and $\mathcal{G}_i^{(2)}$ are the coarser cluster views with approximately $N_i/2$ and $N_i/4$ cluster nodes. For each view $\mathcal{G}_i^{(r)} = (\mathcal{V}_i^{(r)}, \mathcal{E}_i^{(r)}, A_i^{(r)}, X_i^{(r)})$, we use the same encoder-decoder structure to reconstruct node features and compute discrepancy scores.

The encoder aggregates neighborhood information and maps the input node features into latent representations. For the l -th encoder layer, we define

$$H_i^{(r,l+1)} = \sigma \left(\hat{A}_i^{(r)} H_i^{(r,l)} W^{(l)} \right) + R \left(H_i^{(r,l)} \right), \quad (13)$$

where $H_i^{(r,0)} = X_i^{(r)}$, $\hat{A}_i^{(r)}$ is the normalized adjacency matrix of the r -th view, $W^{(l)}$ is the learnable weight matrix, $\sigma(\cdot)$ is the activation function, and $R(\cdot)$ denotes the residual connection. The encoder output is denoted as $Z_i^{(r)}$. The decoder then generates node features from this latent representation:

$$X_i^{gen,(r)} = \text{Decoder} \left(Z_i^{(r)}, A_i^{(r)} \right) \quad (14)$$

If a node follows the common structural pattern of its neighborhood, its generated representation is expected to stay close to the input representation. In contrast, anomalous nodes or anomalous clusters tend to produce larger reconstruction discrepancies.

During training, we optimize the autoencoder over all three graph views. The objective contains a reconstruction discrepancy term and a latent variance regularization term:

$$\mathcal{L} = \frac{1}{3} \sum_{r=0}^2 \left[\mathbb{E} \left(1 - \cos \left(X_i^{(r)}, X_i^{gen,(r)} \right) \right)^\alpha + \beta \mathcal{L}_{var}^{(r)} \right], \quad (15)$$

where $\alpha > 1$ controls the sensitivity of the reconstruction discrepancy, and $\beta > 0$ controls the strength of the variance regularization. The variance loss is defined as

$$\mathcal{L}_{var}^{(r)} = \frac{1}{N_i^{(r)} d'} \sum_{n=1}^{N_i^{(r)}} \sum_{d=1}^{d'} \left(Z_{i,n,d}^{(r)} - \frac{1}{N_i^{(r)}} \sum_{n'=1}^{N_i^{(r)}} Z_{i,n',d}^{(r)} \right)^2, \quad (16)$$

where $N_i^{(r)} = |\mathcal{V}_i^{(r)}|$ is the number of nodes in the r -th graph view, d' is the unified feature dimension, and $Z_{i,n,d}^{(r)}$ is the d -th latent feature of node n . This regularization encourages the latent representations to remain stable and reduces the influence of noisy local variations.

At inference time, we compute an anomaly score for each node in each graph view by measuring the cosine distance between its input feature and generated feature:

$$s_i^{(r)}(u) = 1 - \cos \left(x_{i,u}^{(r)}, x_{i,u}^{gen,(r)} \right) = 1 - \frac{x_{i,u}^{(r)} \cdot x_{i,u}^{gen,(r)}}{\|x_{i,u}^{(r)}\| \|x_{i,u}^{gen,(r)}\|} \quad (17)$$

Here, $x_{i,u}^{(r)}$ and $x_{i,u}^{gen,(r)}$ denote the input and generated features of node u in view r , respectively. A larger value of $s_i^{(r)}(u)$ indicates a stronger discrepancy and therefore a higher anomaly likelihood.

Since the coarser graph views contain cluster nodes, their scores are mapped back to the original nodes. Let $c_r(v)$ denote the node or cluster in view r that contains the original node $v \in \mathcal{V}_i$. The final anomaly score of v is computed by max aggregation:

$$S_i(v) = \max_{r \in \{0,1,2\}} s_i^{(r)}(c_r(v)) \quad (18)$$

This aggregation keeps the strongest anomaly evidence across different views. Therefore, a node can receive a high final score either because it is individually abnormal in the finest view, or because it belongs to a suspicious cluster in a coarser view. During evaluation, $S_i(v)$ is compared with the original node-level ground-truth label, so no additional cluster-level labels are required.

5 Experiments

5.1 Experimental Setup

Datasets. We use twelve datasets from different distributions and domains. The training datasets include ACM, BlogCatalog, Flickr, and Facebook. The test datasets include Cora, CiteSeer, PubMed, Reddit, CS, Photo, Amazon, and YelpChi. More details about the datasets are in Table 1.

Competing Methods. We compare AlignGAD with eleven competing GAD methods, including four supervised methods (GCN [13], GAT [24], GHRN [7], and NRGL [26]), five unsupervised methods (AnomalyDAE [6], DOMINANT [4], TAM [20], GADAM [3], and GGAD [22]), and two generalist methods (UN-Prompt [17] and ARC [14]). All methods are trained and evaluated on the same training and test datasets.

Table 1. The statistics of datasets.

Dataset	Train	Test	#Nodes	#Edges	#Features	%Anomaly
Cora	-	✓	2,708	5,429	1,433	5.53
CiteSeer	-	✓	3,327	4,732	3,703	4.50
ACM	✓	-	16,484	71,980	8,337	3.62
PubMed	-	✓	19,717	44,338	500	3.04
BlogCatalog	✓	-	5,196	171,743	8,189	5.77
Flickr	✓	-	7,575	239,738	12,047	5.94
Facebook	✓	-	1,081	55,104	576	2.31
Reddit	-	✓	10,984	168,016	64	3.33
CS	-	✓	18,333	81,894	6,805	3.27
Photo	-	✓	7,650	119,081	745	5.88
Amazon	-	✓	10,244	175,608	25	6.76
YelpChi	-	✓	23,831	49,315	32	5.10

Experiment Setting. All baselines are implemented using their codebases with default optimization and hyperparameter settings under the same zero-shot protocol. In AlignGAD, we use SVD to project node features of all graphs to an 8-dimensional shared space. The node discrepancy model uses a 3-layer GCN encoder and a 2-layer GCN decoder, with a linear transformation layer between them to improve representation capacity. We train the model on the source graphs and directly evaluate it on all unseen target graphs without fine-tuning.

Evaluation Metrics. We evaluate all methods using Area Under the Receiver Operating Characteristic Curve (AUROC) and Area Under the Precision-Recall Curve (AUPRC). Higher AUROC and AUPRC values indicate better anomaly detection performance.

5.2 Performance

Table 2 presents the AUROC and AUPRC results of AlignGAD compared with supervised, unsupervised, and generalist GAD baselines. Based on the results, we highlight the following observations:

- **AlignGAD achieves strong overall performance in the zero-shot setting.** Without target-domain fine-tuning, AlignGAD obtains the best AUROC on six out of eight datasets and the best AUPRC on five datasets. This shows that the proposed framework can learn anomaly-related patterns that remain useful across unseen graphs.
- **Cluster-aware scoring provides useful anomaly evidence beyond individual nodes.** AlignGAD performs particularly well on PubMed, CiteSeer, Photo, Cora, and Reddit, where it achieves the best result in at least one evaluation metric. This suggests that assigning scores through cluster-level graph views helps reveal abnormal patterns that may be weak at the individual node level.

Table 2. AUROC and AUPRC results compared with baselines on 8 real-world datasets. For each dataset, the best performance within each metric is boldfaced.

Metric	Type	Method	PubMed	CS	CiteSeer	Photo	Amazon	Cora	Reddit	YelpChi
AUROC	Supervised	GCN (ICLR'17)	0.537	0.544	0.667	0.533	0.503	0.559	0.498	0.465
		GAT (ICLR'18)	0.427	0.385	0.399	0.488	0.489	0.524	0.518	0.518
		GHRN (WebConf'23)	0.378	0.348	0.619	0.585	0.456	0.548	0.505	0.501
		NRGL (IJCAI'24)	0.362	0.568	0.573	0.491	0.439	0.562	0.470	0.474
	Unsupervised	AnomalyDAE (ICASSP'20)	0.604	0.619	0.484	0.530	0.308	0.530	0.459	0.407
		DOMINANT (SDM'19)	0.391	0.500	0.543	0.390	0.497	0.548	0.484	0.473
		TAM (NeurIPS'23)	0.664	0.630	0.557	0.495	0.601	0.437	0.502	0.486
		GADAM (ICLR'24)	0.462	0.497	0.587	0.442	0.564	0.503	0.453	0.462
		GGAD (NeurIPS'24)	0.289	0.412	0.252	0.498	0.388	0.295	0.519	0.513
	Generalist GAD	UNPrompt (IJCAI'25)	0.654	0.584	0.607	0.487	0.430	0.627	0.508	0.444
		ARC (NeurIPS'24)	0.557	0.629	0.574	0.463	0.456	0.613	0.481	0.482
		AlignGAD (Ours)	0.807	0.764	0.799	0.590	0.603	0.750	0.601	0.458
AUPRC	Supervised	GCN (ICLR'17)	0.033	0.036	0.078	0.063	0.068	0.063	0.033	0.047
		GAT (ICLR'18)	0.024	0.024	0.035	0.053	0.066	0.053	0.037	0.051
		GHRN (WebConf'23)	0.026	0.073	0.057	0.063	0.063	0.067	0.043	0.066
		NRGL (IJCAI'24)	0.064	0.083	0.053	0.057	0.115	0.044	0.038	0.069
	Unsupervised	AnomalyDAE (ICASSP'20)	0.045	0.184	0.040	0.077	0.044	0.056	0.030	0.042
		DOMINANT (SDM'19)	0.021	0.038	0.030	0.036	0.056	0.051	0.038	0.049
		TAM (NeurIPS'23)	0.095	0.044	0.052	0.054	0.062	0.046	0.033	0.057
		GADAM (ICLR'24)	0.102	0.062	0.072	0.081	0.074	0.052	0.036	0.055
		GGAD (NeurIPS'24)	0.019	0.073	0.030	0.079	0.050	0.035	0.033	0.061
	Generalist GAD	UNPrompt (IJCAI'25)	0.074	0.048	0.062	0.057	0.017	0.083	0.034	0.043
		ARC (NeurIPS'24)	0.071	0.197	0.072	0.055	0.018	0.099	0.030	0.048
		AlignGAD (Ours)	0.241	0.322	0.297	0.0931	0.030	0.224	0.045	0.046

- **AlignGAD reduces the dependence on dataset-specific supervision.** Supervised baselines can perform well on some target graphs, such as GAT on YelpChi and NRGL on Amazon, but their behavior is less consistent under domain shift. In contrast, AlignGAD relies on feature alignment and reconstruction discrepancy rather than target labels, making it more suitable for the generalized GAD scenario.

5.3 Ablation Study

To examine the contribution of each component in AlignGAD, we conduct an ablation study on four representative test datasets. We consider three variants: w/o Global Unification, which removes the whole global unification process; w/o Spectral Norm., which keeps feature dimension alignment but removes spectral domain normalization; and w/o Clustering, which uses only the original graph without cluster-aware graph views. As shown in Table 3, removing the Global Unification Module generally leads to performance degradation on PubMed, CiteSeer, Photo, and Cora, indicating that aligning heterogeneous graph features is important for reducing cross-domain discrepancy. The drop becomes more significant when spectral normalization is removed, especially on PubMed and CiteSeer, showing that frequency-domain normalization helps produce more stable representations across unseen graphs. The w/o Clustering variant also performs worse than the full model on PubMed, CiteSeer, and Photo, which confirms that cluster-aware graph views provide useful group-level information beyond individual node reconstruction. Although w/o Spectral Norm. achieves a slightly higher result on Cora, the full AlignGAD model performs more consistently across datasets. Overall, these results show that global unification, spectral

Table 3. Ablation study on the key components of AlignGAD. The term “w/o” indicates “without”.

Variation	PubMed	CiteSeer	Photo	Cora
AlignGAD	0.7560	0.6990	0.6790	0.6300
w/o Global Unification	0.7499	0.6807	0.6559	0.5887
w/o Spectral Norm.	0.5882	0.5969	0.6433	0.6573
w/o Clustering	0.6819	0.6300	0.6151	0.5984

normalization, and clustering play complementary roles in improving zero-shot graph anomaly detection.

6 Conclusion

In this paper, we propose AlignGAD, a zero-shot generalized graph anomaly detection framework for heterogeneous graphs. AlignGAD learns transferable anomaly patterns from multiple source graphs and directly applies them to unseen target graphs without fine-tuning. By combining graph feature alignment, cluster-aware graph views, and reconstruction-based discrepancy scoring, the model captures both individual node deviations and group-level abnormal patterns. Experiments on multiple real-world datasets demonstrate the effectiveness of AlignGAD under the zero-shot GAD setting. In future work, we plan to further improve the clustering strategy and explore more adaptive aggregation mechanisms for diverse graph domains.

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